

CSE340 Assignment

Am. to the Ques. No. 1

a) Yield is the ρ ratio of successful and functional dies produced on a silicon wafer relative to the maximum possible number of dies on that wafer.

Wafer production involves hundred of highly complex, expensive steps, the processing cost per wafer is fixed. A high yield directly increases the number of sellable chips, dramatically lowering the cost per die and determining whether a manufacturing process is financially viable.

b) ' $\approx$ ' is used instead of ' $=$ ', because Dies per wafer cannot be a float number. Silicon wafers are circular shaped and dies are rectangular, that's we cannot utilize the 100% of the wafer. So we have to floor the value of dies per wafer to come to realistic solution.

c) We know,

$$\text{Yield} = \frac{1}{(1 + (\text{Defects per area} \times \text{die area}))^{0.5}}$$

For wafer A,

$$\text{Yield}_A = \frac{1}{(1 + (0.025 \times 3.46))^{0.5}}$$

$$= 0.75$$

For wafer B,

$$\text{Yield}_B = \frac{1}{(1 + (0.018 \times 3.78))^{0.5}}$$

$$= 0.79$$

$$\text{die area}_A = \frac{\text{wafer area}}{110}$$

$$= \frac{2\pi r^2}{110}$$

$$= \frac{\pi \left(\frac{22}{2}\right)^2}{110}$$

$$= 3.46$$

$$\text{die area}_B = \frac{\pi \left(\frac{25}{2}\right)^2}{130}$$

$$= 3.78$$

d) For A,

$$\text{Dies per wafer} \approx \frac{\pi \left(\frac{22}{2}\right)^2}{3.46} = 110$$

$$\text{Cost per die} = \frac{30}{110 \times 0.75} = 0.36 \text{ USD}$$

For B,

$$\text{Cost per wafer} = \frac{28}{130 \times 0.79} = 0.27 \text{ USD.}$$

Ans. to the Ques. No. 2

a) For Alpha,

$$\begin{aligned} \text{average CPI} &= \frac{6 \times 80 \times 10^4 + 3 \times 60 \times 10^4 + 2 \times 40 \times 10^4 + 5 \times 20 \times 10^4}{2 \times 10^6} \\ &= 4.2 \end{aligned}$$

For Beta,

$$\begin{aligned} \text{average CPI} &= \frac{4 \times 80 \times 10^4 + 2 \times 60 \times 10^4 + 2 \times 40 \times 10^4 + 2 \times 20 \times 10^4}{2 \times 10^6} \\ &= 2.8 \end{aligned}$$

b) For Alpha,

$$\text{Clock cycle time} = \frac{1}{2.4 \times 10^9} = 4.17 \times 10^{-10}$$

$$\begin{aligned} \text{execution time} &= 2 \times 10^6 \times 4.2 \times 4.17 \times 10^{-10} \\ &= 3.5 \times 10^{-3} \text{ s} \end{aligned}$$

For Beta,

$$\text{execution time} = \frac{2 \times 10^6 \times 2.8}{3.2 \times 10^9} = 1.75 \times 10^{-3} \text{ s}$$

$$\text{c) SPEC ratio for Alpha} = \frac{90}{3.5} = 25.7$$

Am. to the Ques. No. 3

$$\begin{aligned} \text{MIPS} &= \frac{\text{Instruction Count}}{\text{Execution time} \times 10^6} = \frac{\text{Instruction count}}{\frac{\text{Instruction count} \times \text{CPI}}{\text{Clock rate}} \times 10^6} \\ &= \frac{\text{Clock rate}}{\text{CPI} \times 10^6} \end{aligned}$$

$$\text{a) C1} \rightarrow \text{MIPS} = \frac{2.8 \times 10^9}{1.2 \times 10^6} = 2333.33$$

$$\text{C2} \rightarrow \text{MIPS} = \frac{3.5 \times 10^9}{1.8 \times 10^6} = 1944.44$$

$$\text{C3} \rightarrow \text{MIPS} = \frac{2.2 \times 10^9}{0.9 \times 10^6} = 2444.44$$

$\therefore$ C3, CPU has the highest performance in MIPS.

b) Given, execution time = 8 s

we know,

$$\text{execution time} = \frac{\text{Instruction count} \times \text{CPI}}{\text{Clock rate}}$$

$$\Rightarrow \text{Instruction count} = \frac{\text{execution time} \times \text{clock rate}}{\text{CPI}}$$

$$= \frac{8 \times 2.8 \times 10^9}{1.2} \quad [\text{Using C1}]$$

$$= 1.867 \times 10^{10}$$

Now,

$$c1 \rightarrow \text{clock cycles} = 1.867 \times 10^{10} \times 1.2 = 2.2404 \times 10^{10}$$

$$c2 \rightarrow \text{clock cycles} = 1.867 \times 10^{10} \times 1.8 = 3.36 \times 10^{10}$$

$$c3 \rightarrow \text{clock cycles} = 1.867 \times 10^{10} \times 0.9 = 1.68 \times 10^{10}$$

c)

$$\text{Clock rate}_{\text{old}} = \frac{\text{execution time}_{\text{old}}}{\text{Instruction count}}$$

$$c) \text{ Clock rate}_{\text{old}} = \frac{\text{Instruction Count} \times \text{CPI}_{\text{old}}}{\text{execution time}_{\text{old}}}$$

$$\text{Clock rate}_{\text{new}} = \frac{\text{Instruction count} \times 1.15 \times \text{CPI}_{\text{old}}}{0.75 \times \text{execution time}_{\text{old}}}$$

$$= 1.53 \times \text{clock rate}_{\text{old}}$$

$$C1 \rightarrow \text{Clock rate}_{\text{new}} = 1.53 \times 2.8 = 4.29 \text{ GHz}$$

$$C2 \rightarrow \text{Clock rate}_{\text{new}} = 1.53 \times 3.5 = 5.355 \text{ GHz}$$

$$C3 \rightarrow \text{Clock rate}_{\text{new}} = 1.53 \times 2.2 = 3.366 \text{ GHz}$$

Ans. to the Ques. No. 4

a) We know,

Amdahl's Law,

$$T_{\text{improved}} = \frac{T_{\text{affected}}}{\text{improvement factor}} + T_{\text{unaffected}}$$

$$\text{Now, } T_{\text{total}} = \frac{100}{1.25} + 200 = 280$$

$$b) \quad 300 \times 0.8 = \frac{20080}{N} + 100220$$

$$\Rightarrow 240 = \frac{20080}{N} + 100220$$

$$\Rightarrow \frac{20080}{N} = 140$$

$$\Rightarrow N = \frac{20080}{140} = 143.428$$

$$\therefore N = 143.428$$

By improving 100% of logical and in AND instructions total time can be reduce to 20%.

Am. to the Ques. No. 5

$$a) A \rightarrow \text{MIPS} = \frac{5 \times 10^9}{1 \times 10^6} = 5 \times 10^3$$

$$b) B \rightarrow \text{MIPS} = \frac{3.2 \times 10^9}{0.7 \times 10^6} = 4.57 \times 10^3$$

b) Processor A,

$$\begin{aligned}\text{execution time} &= \frac{8 \times 10^9 \times 1}{5 \times 10^9} \\ &= 1.6 \text{ s}\end{aligned}$$

Processor B,

$$\begin{aligned}\text{execution time} &= \frac{4 \times 10^9 \times 0.7}{3.2 \times 10^9} \\ &= 0.875 \text{ s}\end{aligned}$$

$\therefore$ Processor B is faster.

c) MIPS does not reflect real performance here.

According to MIPS, processor A is better than processor B. But in reality, processor B executes a task almost twice as faster than processor A. Hence processor A needs more instructions for a task than processor B, MIPS does not consider it.

Am. to the Ques. No. 6

We know,

$$\begin{aligned}\text{Power} &= \text{Capacitive load} \times \text{voltage}^2 \times \text{Frequency} \\ &= C V^2 F\end{aligned}$$

$$\begin{aligned}\text{a) Power}_{\text{new}} &= (0.9 \times C) \times (0.85 \times V)^2 \times (0.9 \times F) \\ &= 0.81 \times 0.7225 C V^2 F \\ &= 0.585 C V^2 F \\ &= 0.585 \text{ Power}\end{aligned}$$

$$\begin{aligned}\therefore \text{The } \cancel{\text{power}} \text{ percentage reduction} &= (1 - 0.585) \\ &= 0.415 \\ &= 41.5\%\end{aligned}$$

b) If $\text{Power} = 100 \text{ W}$

$$\begin{aligned}\therefore \text{Power}_{\text{new}} &= \cancel{0.58} 0.585 \times \text{Power} \\ &= 0.585 \times 100\end{aligned}$$

$$\frac{5 \times 0.1 + 2 \times 1}{0.1 + 2.1 + 8 + 10} = \frac{58.5 \text{ W} + 12 \times 2}{10.2 + 10} = 190 \text{ p.u.}$$

c) Reducing voltage more impactful than reducing frequency because voltage is quadratically proportional to power but on the other hand frequency has linear relationship with power.

Ans. to the Ques. No. 7

a) Given, clock period = $50 \text{ ps} = 50 \times 10^{-12} \text{ s}$

$$\therefore \text{frequency} = \frac{1}{50 \times 10^{-12}} = 20 \text{ GHz}$$

b) CPI —

$$\text{Add} = 2$$

$$\text{Sub} = 1$$

$$\text{Mul} = 1.5$$

$$\text{Div} = 2$$

$$\text{c) Avg. CPI} = \frac{2 \times 21 + 3 \times 1 + 15 \times 1.5 + 10 \times 2}{21 + 3 + 15 + 10}$$

$$= 1.79$$

$$d) \text{ Execution time} = \frac{(21+3+15+10) \times 1.79}{20 \times 10^9}$$

$$= 4.38 \times 10^{-9} \text{ s}$$

$$e) \text{ SPEC ratio} = \frac{1.08 \times 10^{-9}}{4.38 \times 10^{-9}}$$

$$= 0.247$$

f) We know,

$$\text{total execution time} = 4.38 \times 10^{-9} \text{ s}$$

execution time for add operations =

$$= 21 \times 2 \times 50 \times 10^{-12}$$

$$T_{\text{affected}} = 2.1 \times 10^{-9} \text{ s}$$

$$\therefore T_{\text{unaffected}} = 4.38 \times 10^{-9} - 2.1 \times 10^{-9}$$

$$= 2.28 \times 10^{-9}$$

Now,

$$T_{\text{improved}} = \frac{T_{\text{affected}}}{N} + T_{\text{unaffected}}$$

$$\Rightarrow 1.2 \times 4.38 \times 10^{-9} = \frac{2.1 \times 10^{-9}}{N} + 2.28 \times 10^{-9}$$

$$\Rightarrow 5.25 \times 10^{-9} = \frac{2.1 \times 10^{-9}}{N} + 2.28 \times 10^{-9}$$

$$\Rightarrow 2.976 \times 10^{-9} = \frac{2.1 \times 10^{-9}}{N}$$

$$\Rightarrow N = 0.7$$

$\therefore$ Add operation must be improved by 0.7.

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$$4.38 \times 10^{-9} = \text{X}$$

$$T_{\text{improved}} = \frac{2.1 \times 10^{-9}}{1.2} + 2.28 \times 10^{-9}$$

$$= 1.75 \times 10^{-9} + 2.28 \times 10^{-9}$$

$$= 4.03 \times 10^{-9} \text{ s}$$

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